

FocusFusion: Cross-Attention Fusion of LiDAR and Vision Features for 3D Semantic Segmentation



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Introduction

Some block contents, followed by a diagram, followed by a dummy paragraph.

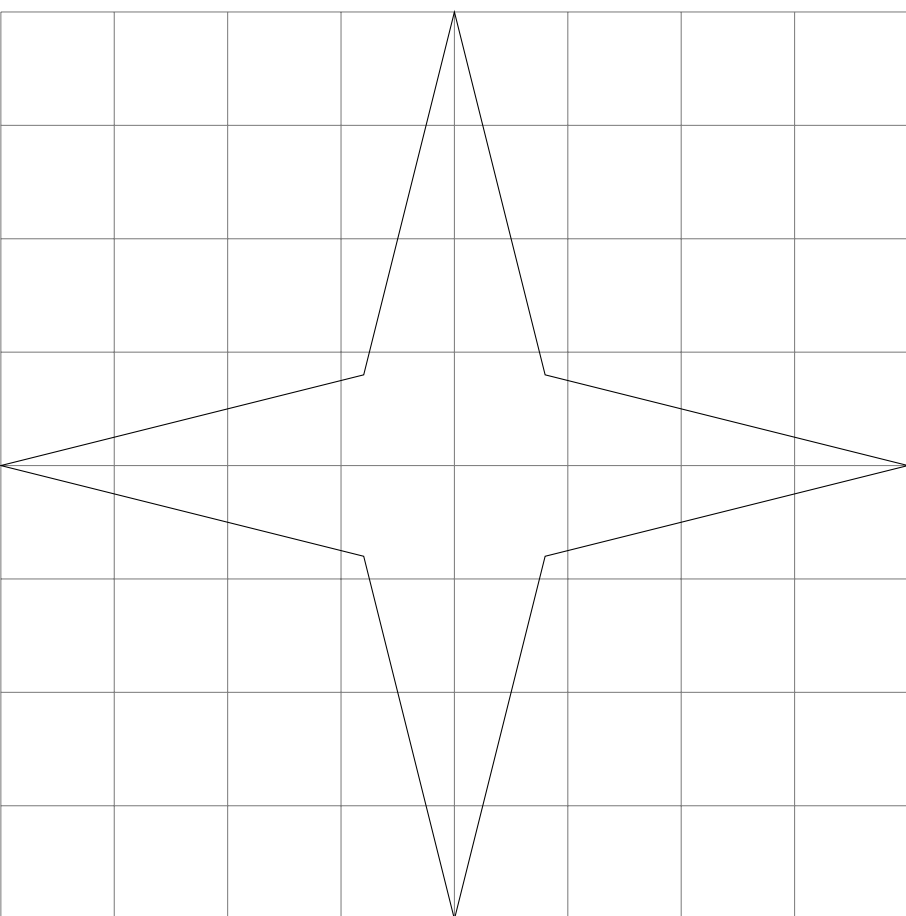


Figure 1. A figure caption.

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Problem Statement

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- **Mauris tempor** risus nulla, sed ornare
- **Libero tincidunt** a dui congue vitae
- **Dui ac pretium** morbi justo neque, ullamcorper

Eget augue porta, bibendum venenatis tortor.

Dataset

A highlighted block

This block catches your eye, so **important stuff** should probably go here.

Curabitur eu libero vehicula, cursus est fringilla, luctus est. Morbi consectetur mauris quam, at finibus elit auctor ac. Aliquam erat volutpat. Aenean at nisl ut ex ullamcorper eleifend et eu augue. Aenean quis velit tristique odio convallis ultrices a ac odio.

- **Fusce dapibus tellus** vel tellus semper finibus. In consequat, nibh sed mattis luctus, augue diam fermentum lectus.
- **In euismod erat metus** non ex. Vestibulum luctus augue in mi condimentum, at sollicitudin lorem viverra.
- **Suspendisse vulputate** mauris vel placerat consectetur. Mauris semper, purus ac hendrerit molestie, elit mi dignissim odio, in suscipit felis sapien vel ex.

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Methods

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Experiments & Analysis

Method	mIoU	mAcc	fwIoU
LitePT	0.606	0.808	0.911
FocusFusion	0.810	0.876	0.903

Table 1. Baseline: LitePT vs. FocusFusion. **+20.4% mIoU** from fusing visual patch features, which help disambiguate geometry-ambiguous classes (e.g. poles vs. pedestrians look nearly identical in LiDAR alone).

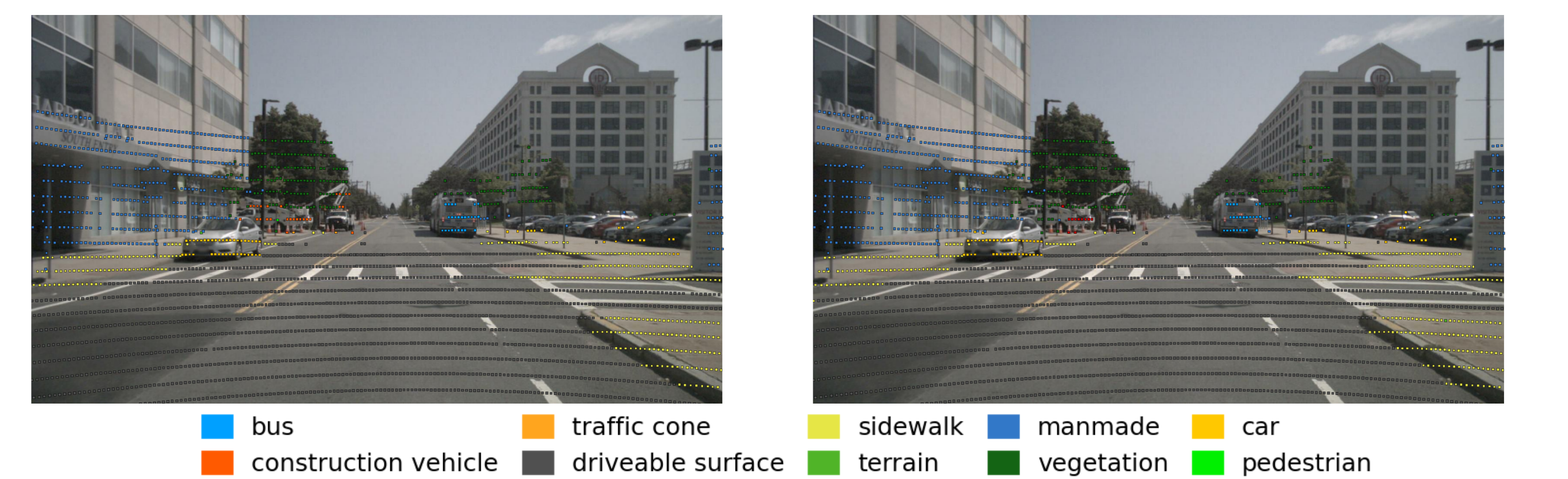


Figure 2. Ground Truth (left) vs. FocusFusion Prediction (right). LiDAR points projected onto the front camera and colored by semantic class. Road, vegetation, and vehicles are well-segmented; main failures are sidewalk/terrain confusion and missed traffic cones.

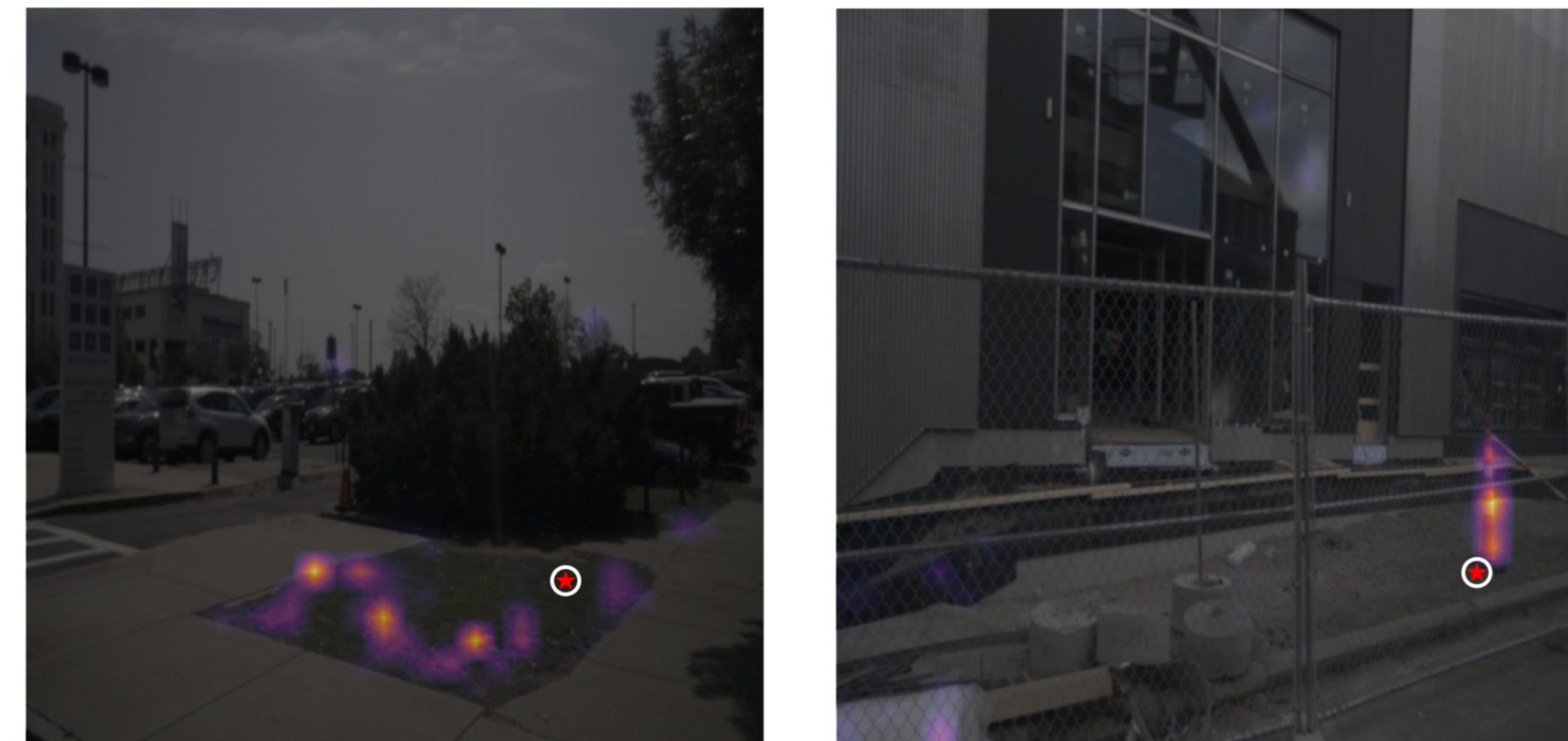


Figure 3. Cross-attention maps: terrain query (left), traffic cone query (right). Warmer colors indicate higher attention weight. Terrain: diffuse attention over grass. Traffic cone: sharp spike on orange patches. Correspondences learned **without any calibration**.

Ablation Studies

Features	mIoU	mAcc	fwIoU
L12 only ($D_v=384$)	0.683	0.798	0.909
L9+L12 ($D_v=768$)	0.685	0.782	0.907

Table 2. DINOv2 layer selection. Negligible gain from doubling D_v ; L12 alone is sufficient at this data scale.

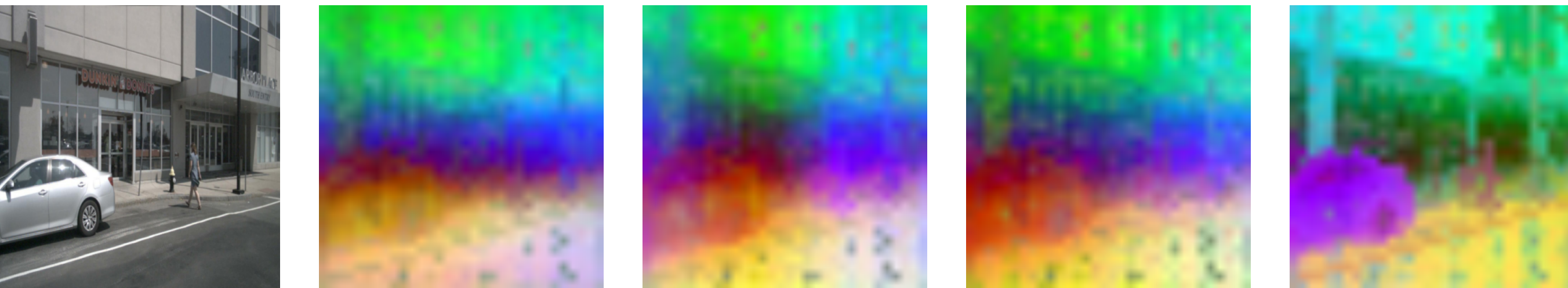


Figure 4. PCA of DINOv2 patch embeddings (raw image, L9–L12). Deeper layers resolve increasingly fine object boundaries.

λ	mIoU	mAcc	fwIoU
0	0.639	0.809	0.903
0.25	0.683	0.798	0.909
0.5	0.681	0.783	0.910

Table 3. Lovász loss weight λ . $\lambda=0.25$ optimal; pure CE biases toward common classes and $\lambda=0.5$ over-penalizes, degrading mAcc.

T	mIoU	mAcc	fwIoU
1	0.810	0.876	0.903
6 (3 s)	0.806	0.875	0.902

Table 4. Temporal memory bank depth T . $T=1$ wins at this data scale; longer K/V sequences need more training diversity to supervise.

Conclusions & Future Work

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First column	Second column	Third column	Fourth
Foo	13.37	384,394	α
Bar	2.17	1,392	β
Baz	3.14	83,742	δ
Qux	7.59	974	γ

Table 5. A table caption.

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References